

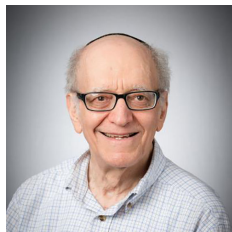
WhileCC-approximability and Acceptability of Elementary Functions or The story of the time I invented a different kind of Wheel

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CiE 2026
Trier University
July 27th - 31st, 2027



In loving memory of Prof. Jeff Zucker



Overview

- Problem
- Key Insights from Original Proof
- A Simpler Approach

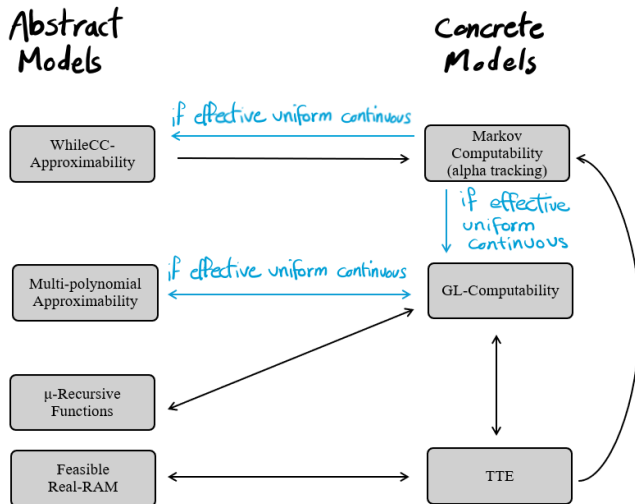
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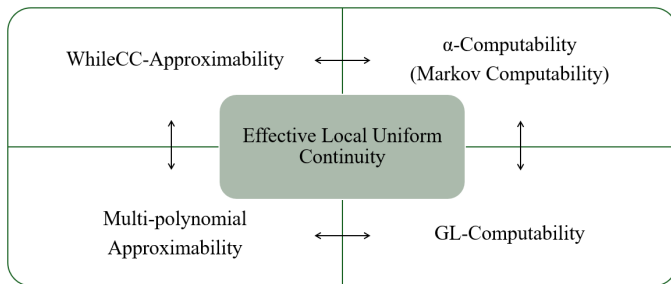
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Computability of Total Functions on $\mathbb{R}$



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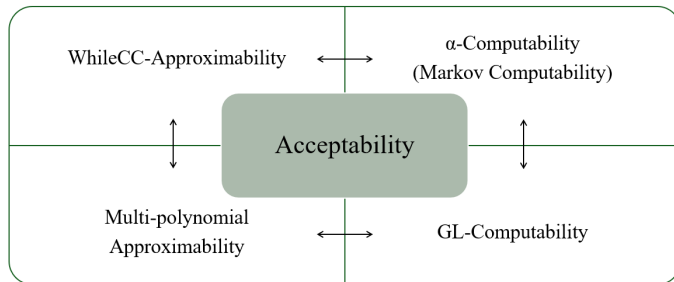


Computability of Partial Functions on $\mathbb{R}$

For partial functions on $\mathbb{R}$, Fu and Zucker [2014] generalize effectively locally uniform continuity to **acceptability** to get an equivalence.

Problem

How general is this class of acceptable functions?



Useful First Step Towards Solution

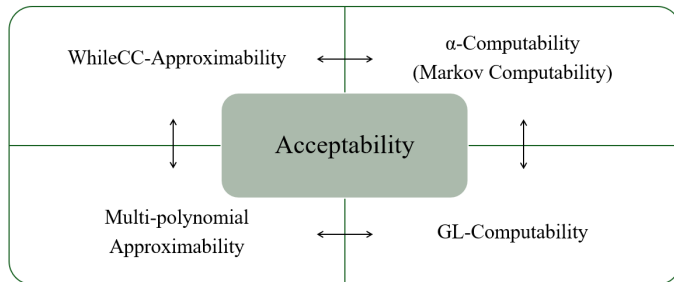
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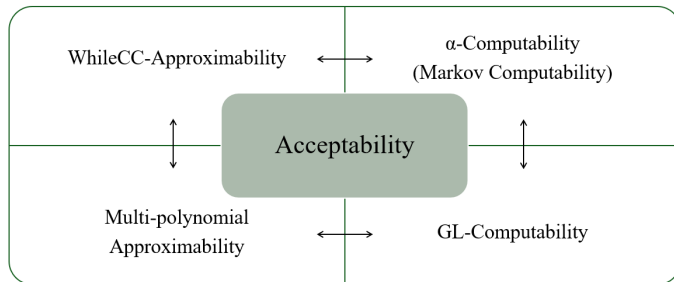
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Definition (**Acceptability**)

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is **acceptable** if there exists a sequence X where:

- 1 X is an **effective open exhaustion** for $\text{dom}(f)$, and
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Background – Elementary Functions

Definition ([Tenenbaum and Pollard, 1985])

The **elementary functions** on $\mathbb{R}$ are partial functions defined by expressions built up from

- computable reals, and
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by applying (repeatedly) the basic operations below on elementary functions f, g :

- $(f + g)(x) = f(x) + g(x)$
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The domains of elementary functions are not all open!

Solution: Modifications

- We define $\sqrt[n]{x} = 0$ for $x < 0$ when n is even.
- We extend the definition of $\arcsin(x)$ to be $\frac{\pi}{2}$ for $x > 1$ and to be $-\frac{\pi}{2}$ for $x < -1$.

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The domains of elementary functions are not all open!

Solution: Modifications

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Definition ([Tenenbaum and Pollard, 1985])

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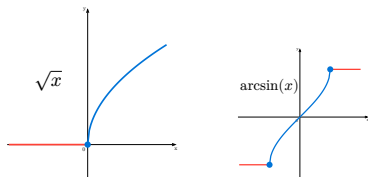
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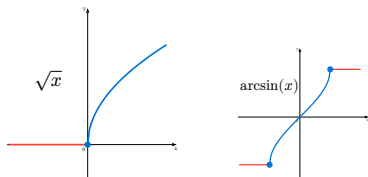
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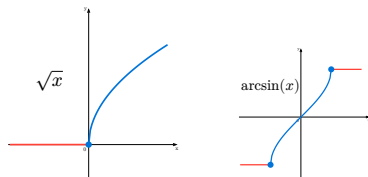
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Contributions

Recall: Equivalence Theorem, [Fu and Zucker, 2014]

For any **acceptable function** $f : \mathbb{R} \rightarrow \mathbb{R}$ and any effective open exhaustion X for $\text{dom}(f)$, the following are equivalent:

- f is an α -computable function.
- f is **WhileCC**-approximable.
- f is GL-computable w.r.t. X .
- f is effectively locally uniformly multipolynomially approximable w.r.t. X .

Theorem 1 (**WhileCC**-approximability Theorem)

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Theorem 2.2 (Acceptability Theorem: Part 2)

Any elementary function is effectively locally uniformly continuous w.r.t. an effective open exhaustion for its domain.

Result 1

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Syntax

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Result 1 - Background - WhileCC Programming Language

Syntax

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$$t^s ::= x^s \mid F(t_1^{s_1}, \dots, t_m^{s_m})$$

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Algebra $\mathcal{R}$

$$\begin{aligned} 0_{\mathbb{R}}, 1_{\mathbb{R}}, -1_{\mathbb{R}} &: \rightarrow \mathbb{R} \\ +_{\mathbb{R}}, \times_{\mathbb{R}} &: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R} \\ +_{\mathbb{N}}, \times_{\mathbb{N}} &: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N} \\ \text{inv}_{\mathbb{R}} &: \mathbb{R} \rightarrow \mathbb{R} \\ 0_{\mathbb{N}} &: \rightarrow \mathbb{N} \\ \text{suc}_{\mathbb{N}} &: \mathbb{N} \rightarrow \mathbb{N} \\ \text{tt}, \text{ff} &: \rightarrow \mathbb{B} \\ \text{and}, \text{or} &: \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B} \\ \text{not} &: \mathbb{B} \rightarrow \mathbb{B} \\ =_{\mathbb{N}}, <_{\mathbb{N}} &: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{B} \\ =_{\text{real}}, <_{\mathbb{R}} &: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{B} \end{aligned}$$

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Definition (**WhileCC**-approximability, [Fu and Zucker, 2014])

A **WhileCC**-procedure P of type $\text{real} \times \text{nat} \rightarrow \text{real}$ on $\mathcal{R}$ is said to *approximate* a function $f : \mathbb{R} \rightarrow \mathbb{R}$ iff for all $n \in \mathbb{N}$ and all $x \in \mathbb{R}$:

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where $\text{Nbd}(y, r)$ has the standard definition of neighborhood on $\mathbb{R}$ i.e.,

$$\text{Nbd}(y, r) = \{z \in \mathbb{R} \mid |y - z| < r\}.$$

Goal

Construct **WhileCC**-procedures approximating elementary functions by induction

Challenge: Using comparison operators introduces undefinedness.

How do we **WhileCC**-approximate “piecewise” functions?

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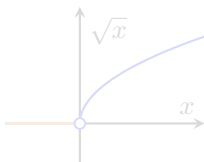
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Result 1 - Challenges

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When defining piecewise functions, comparison makes a hole!

Example: Even Root - First Attempt



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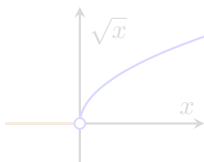
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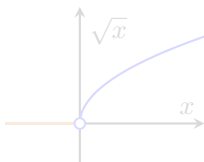
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Example: Even Root - First Attempt



Solution

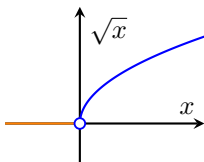
- Find an overlapping interval where both pieces are defined and give a good enough answer
- Use the nondeterminism of “choose”

Result 1 - Challenges

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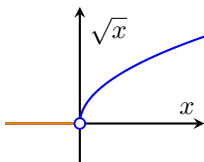
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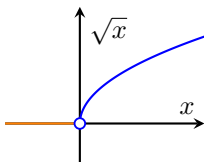
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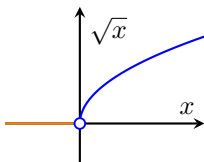
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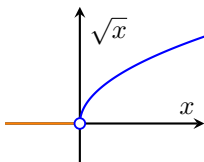
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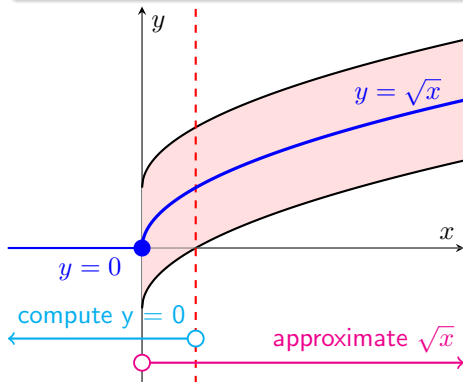
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Result 2

Theorem 2.2 (Acceptability Theorem: Part 2)

Any elementary function is effectively locally uniformly continuous **w.r.t. an effective open exhaustion** for its domain.

(We prove that this is) equivalent to proving:

Any elementary function has a local modulus of continuity.

Definition (Local modulus of continuity)

Let $f : \mathbb{R} \rightarrow \mathbb{R}$. A recursive function $N : \mathbb{Q} \times \mathbb{Q} \times \mathbb{N} \rightarrow \mathbb{N}$ is called a **local modulus of continuity** for f iff for any $a, b \in \mathbb{Q}$ with $[a, b] \subseteq \text{dom}(f)$ and $k \in \mathbb{N}$, we have

$$\forall x, y \in (a, b) \quad |x - y| < 2^{-N(a,b,k)} \implies |f(x) - f(y)| < 2^{-k}.$$

The notion of effective local uniform continuity is independent of effective open exhaustion.

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Summary

We proved that:

- all elementary functions are **WhileCC**-approximable.
- all elementary functions are acceptable.

We also

- presented an **alternative characterization** of acceptable functions using the **local modulus of continuity** concept.

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Alternative Approach

Theorem

The following are true for any function f on the reals:

- f is GL-computable $\Rightarrow f$ is effectively uniformly continuous w.r.t. any compact set [Anonymous Reviewer].
- f is GL-computable + has an r.e. open domain $\Rightarrow f$ is effectively uniformly continuous [Pauly, 2016].

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Elementary functions are GL-computable and have open domains.

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Straightforward proof by induction. □

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Future Work

Conjecture:

- All partial unary **WhileCC**-approximable functions are acceptable.
 - **If the conjecture holds**, are *non-unary* **WhileCC**-approximable functions acceptable?
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References

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A HUGE Thank you!